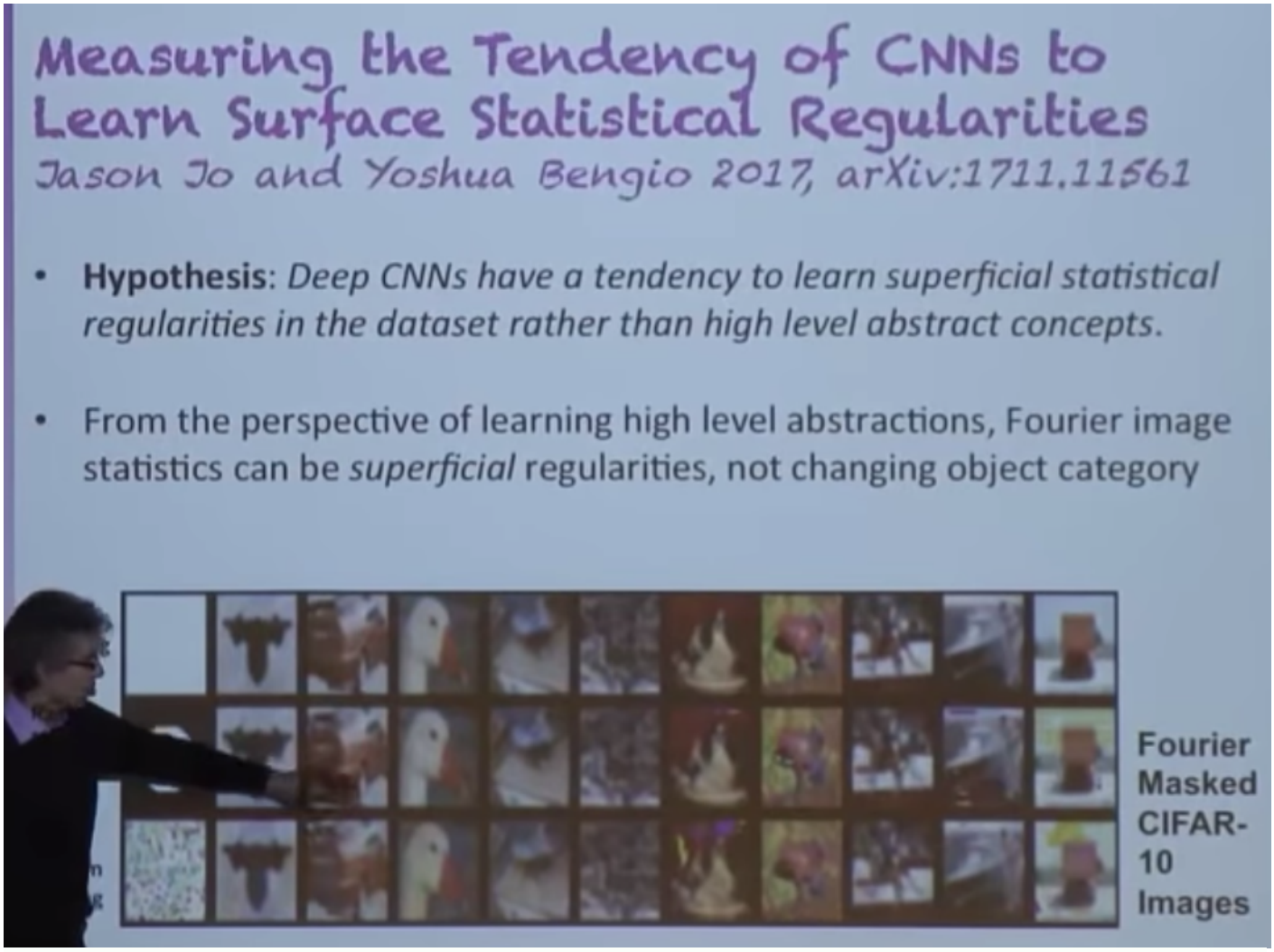
From Deep Learning of Disentangled Representations to Higher-level Cognition - YouTube

Map data from real world to a low-dimensional sparse abstract space



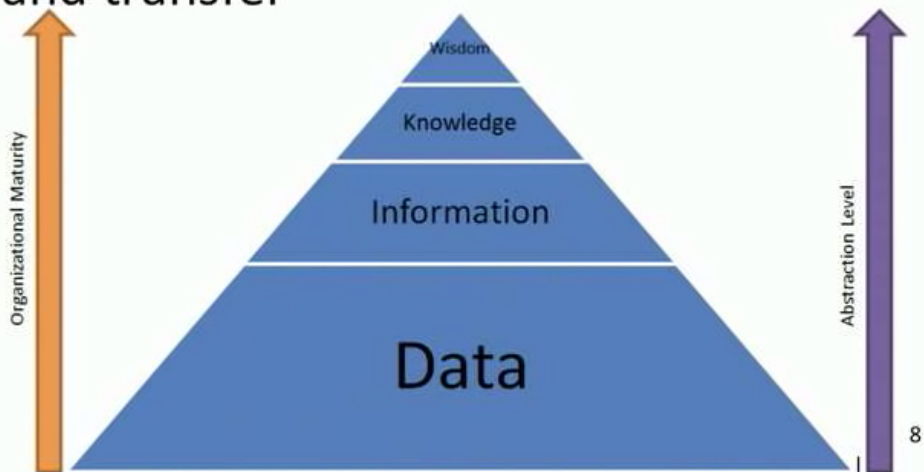
Learning « How the world ticks »

- So long as our machine learning models « cheat » by relying only on superfcial statistical regularities, they remain vulnerable to out-of-distribution examples
- Humans generalize better than other animals thanks to a more accurate internal model of the **underlying causal relationships**
- To predict future situations (e.g., the effect of planned actions) far from anything seen before while involving known concepts, an essential component of reasoning, intelligence and science

Learning Multiple Levels of Abstraction

(Bengio & LeCun 2007)

- The big payoff of deep learning is to allow learning higher levels of abstraction
- Higher-level abstractions **disentangle the factors of variation**, which allows much easier generalization and transfer



8

Invariance and Disentangling

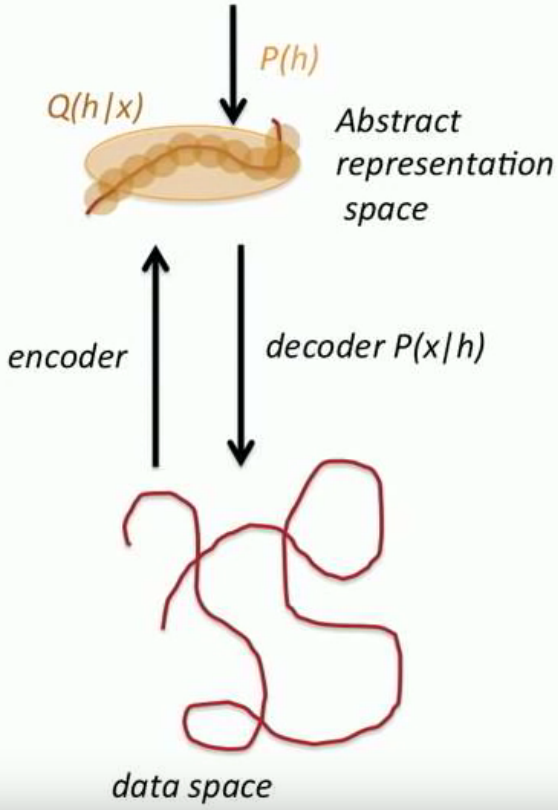
- Invariant features
- Which invariances?

- Alternative: learning to disentangle factors
- Good disentangling →
  avoid the curse of dimensionality

9

Latent Variables and Abstract Representations

- Encoder/decoder view: maps between low & high-levels
- Encoder does inference: interpret the data at the abstract level
- Decoder can generate new configurations
- Encoder flattens and disentangles the data manifold
- Marginal independence in h-space



Independently Controllable Factors

(Emmanuel Bengio, Valentin Thomas, Joelle Pineau, Doina Precup, Yoshua Bengio, 2017)
(Valentin Thomas, Jules Pondard, Emmanuel Bengio, Marc Sarfati, Philippe Beaudoin, Marie-Jean Meurs, Joelle Pineau, Doina Precup, Yoshua Bengio, 2017)

- Jointly train for each aspect (factor)
  - A policy π_k (which tries to selectively change just that factor)
  - A representation (which maps state to value of factor) f_k

Discrete case, $\phi \in \{1, .., N\}$, define *selectivity*:

$$\sum_{k=1}^{N} \mathbb{E}_{(s_t, a_t, s_{t+1})} \left[\pi_k(a_t|s_t) \frac{f_k(s_{t+1}) - f_k(s_t)}{\sum_{k'} |f_{k'}(s_{t+1}) - f_{k'}(s_t)|} \right]$$

- Optimize both policy π_k and representation f_k to minimize

$$\underbrace{\mathbb{E}_s[\tfrac{1}{2}||s - g(f(s))||_2^2]}_{\text{reconstruction error}} - \lambda \underbrace{\sum_k \mathbb{E}_s[\sum_a \pi_k(a|s) \log sel(s, a, k)]}_{\text{disentanglement objective}}$$

Abstraction Challenge for Unsupervised Learning

- Why is modeling P(acoustics) so much worse than modeling P(acoustics | phonemes) P(phonemes)?

- Wrong level of abstraction?
  - many more entropy bits in acoustic details then linguistic content

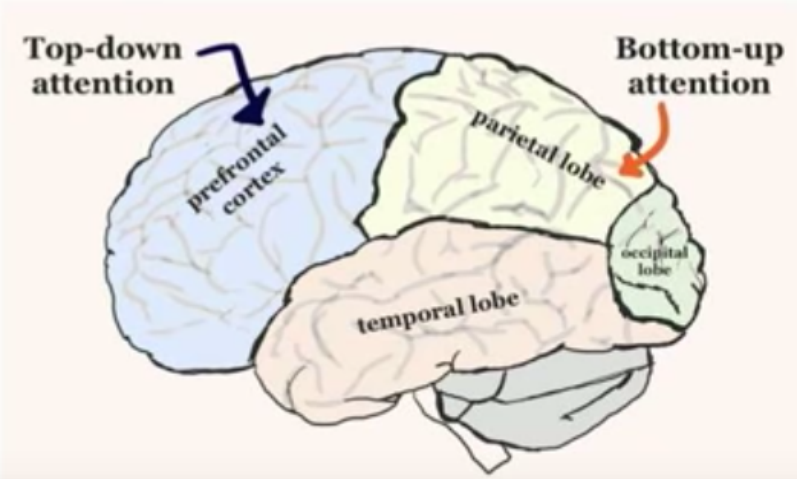
→ predict the future in abstract space instead

On the Relation between Abstraction and Attention

- Attention allows to focus on a few elements out of a large set
- Soft-attention allows this process to be trainable with gradient-based optimization and backprop

Attention focuses on a few appropriate abstract or concrete elements of mental representation

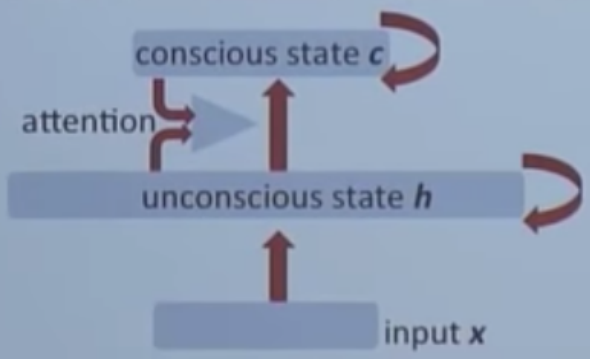
Access consciousness is one aspect of consciousness, what is accessed and comes to mind (Dehaene's C1 or global availability), See *Dehaene et al, Science, 2017*.

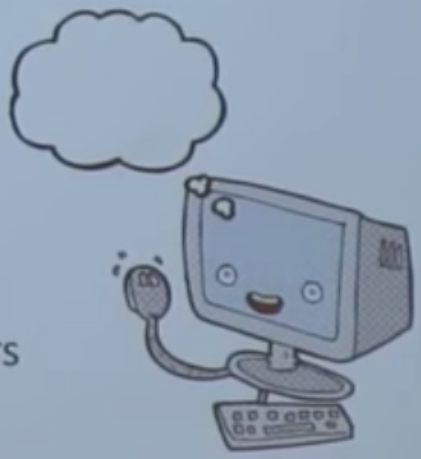


The Consciousness Prior

Bengio 2017, arXiv:1709.08568

- 2 levels of representation:
  - High-dimensional abstract representation space (all known concepts and factors) ***h***
  - Low-dimensional conscious thought ***c***, extracted from ***h***





- ***c*** includes names (keys) and values of factors

32